

Information Geometry

0) Preliminaries

Let M be a manifold. Let $D: M \times M \rightarrow \mathbb{R}_{\geq 0}$. For $p \in M$, let

$$g_p = \partial_p^2 D: T_p M \otimes T_p M \rightarrow \mathbb{R}.$$

Then D is a **divergence** if

$$D(p, q) \geq 0 \quad \forall p, q \in M \quad \text{with } D(p, q) = 0 \text{ iff } p = q,$$

and g_p is positive-definite. note that it is automatically symmetric

Note: We have the following symmetries that follow from $D(p, q) \geq 0$ with equality only on the diagonal.

First order derivatives vanish because $D(p, \cdot)$ and $D(\cdot, p)$ have (global)

(I) minima at p : $i^*(\partial_1 D) = i^*(\partial_2 D) = 0 = i^*(\partial D)$ diagonal embedding $i: M \hookrightarrow M \times M$

For $X \in TM$, let $X_1, X_2 \in TM \times TM$. Then

(II) $i^*(X_1, Y_1, D) = i^*(X_2, Y_2, D) = -i^*(X_1, Y_2, D)$

by differentiating (I). Note that this defines the associated Riemannian metric.

We define the **primal connection** ∇ by

$$g(\nabla_X Y, Z) = -i^*(X_1, Y_1, Z_2, D),$$

and the **dual connection** ∇^* by

$$g(\nabla_X^* Y, Z) = -i^*(X_2, Y_2, Z_1, D).$$

Note: Both the primal and the dual connections are **torsion-free**, i.e.

$$T(X, Y) := \nabla_X^* Y - \nabla_Y^* X - [X, Y] = 0.$$

Note: These connections are dual in the sense of

$$Z g(X, Y) = g(\nabla_Z X, Y) + g(X, \nabla_Z^* Y).$$

In particular, the Levi-Civita connection ∇^{LC} is decomposed as

$$\nabla^{LC} = \frac{1}{2}(\nabla + \nabla^*).$$

Example: Let M be a Riemannian manifold. Then locally, the distance function d is a divergence. It induces the Riemannian metric and its primal and dual connections equal the Levi-Civita connection.

Example: Let $\Psi: \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. Then

$$D_\Psi[\cdot: \cdot] := \Psi(z) - \Psi(z_0) - \nabla \Psi(z_0) \cdot (z - z_0),$$

the distance of the graph of Ψ at z to the hyperplane tangent to the graph at z_0 , is a divergence, called the **Bregman divergence**.

Example (Euclidean): For $\Psi(z) = \frac{1}{2} \sum z_i^2$, the Bregman divergence is the Euclidean distance.

Example (Kullback-Leibler): Let $\Psi(z) = \sum z_i \log z_i$ be the (negative) Shannon entropy. Then the Bregman divergence is the **Kullback-Leibler divergence**

$$D[\cdot: \cdot] = \sum (z_i \log \frac{z_i}{z'_i} - z_i + z'_i)$$

zero for probability measures

Example (Free energy of an exponential family): Let $p: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$ be

$$p(x, \theta) = \exp(\sum \theta_i x_i + k(x) - \Psi(\theta)),$$

such that $\int p(x, \theta) dx = 1$. We view the map $\theta \mapsto p(\cdot, \theta)$ as a coordinate chart on the set of probability distributions. Note that the **free energy**

$$\Psi(\theta) = \log \int \exp(\sum \theta_i x_i + k(x)) dx$$

is convex. Differentiating $\int p(x, \theta) dx = 1$ w.r.t. θ , we get

$$\nabla \Psi(\theta) = \mathbb{E}[x].$$

Hence, Ψ induces the KL divergence

$$D[p(\cdot, \theta): p(\cdot, \theta')] = \int p(x, \theta') \log \frac{p(x, \theta')}{p(x, \theta)} dx.$$

Let $\Psi: \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. Consider Ψ as a function on M . Let $\partial \Psi: M \rightarrow T^*M$ have image $\Omega = \text{Im}(\partial \Psi) \subset T^*M$. Then the map

$$\mathcal{L}_\Psi = \partial \Psi: M \xrightarrow{\sim} \Omega$$

is the **Legendre transformation** of Ψ . In local coordinates,

$$(z_1, \dots, z_n) \mapsto (\frac{\partial \Psi}{\partial z_1}, \dots, \frac{\partial \Psi}{\partial z_n}).$$

The Legendre transform is a bijection between the graph of Ψ and its tangent hyperplanes.

z -axis intercept of hyperplane defined by z^* , in particular, $\Psi^*(z^*) = \max_z \{g(z, z^*) - \Psi(z)\}$

We a function $\Psi^*: \Omega \rightarrow \mathbb{R}$ by

$$\Psi^*(z^*) = g(z, z^*) - \Psi(z),$$

The function Ψ can be recovered from the set of tangent hyperplanes together with their z -axis intercepts Ψ^*

where $z = \mathcal{L}_\Psi^{-1}(z^*)$.

Note ($\Psi^{**} = \Psi$): We have

$$\partial \Psi^*(v) = g(\mathcal{L}_\Psi^{-1}(v), z^*) + g(z, \partial \Psi^*(v)) - \partial \Psi(v) = g(z, v),$$

i.e. $\partial \Psi^* = z = \mathcal{L}_\Psi^{-1}(z^*)$.

Note: The Bregman divergence of Ψ^* equals that of Ψ .

$$D_{\Psi^*}[z^*: z'^*] = D_\Psi[z: z'].$$

Note: We have

$$D_\Psi[P: Q] = \Psi(z_P) + \Psi^*(z_Q^*) - z_P \cdot z_Q^*$$

Let ∇, ∇^* be dual connections on a Riemannian manifold M , i.e.

$$X g(Y, Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X^* Z).$$

We call (M, ∇, ∇^*) **dually flat** if the Riemann curvature of ∇ (and equivalently of ∇^*) vanishes:

$$R(X, Y)Z := \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X, Y]} Z = 0$$

Note: Generally, we have

$$g(R(X, Y)Z, W) = -g(Z, R^*(X, Y)W).$$

Note: We calculate

$$\begin{aligned} (\nabla_X g)(Y, Z) &= X g(Y, Z) - g(\nabla_X Y, Z) - g(Y, \nabla_X^* Z) \\ &= g(Y, \nabla_X^* Z - \nabla_X^* Z) =: g(Y, K(X, Z)) \\ &= (\nabla_X g)(Y, Z), \end{aligned}$$

symmetric because ∇, ∇^* are torsion-free

i.e. ∇g is a **Codazzi tensor**.

Dual flatness is equivalent to a local triviality. To prove it, we recall the following

Theorem (Poincaré lemma for flat vector bundles): Let $(E, \nabla) \rightarrow M$ be a flat vector bundle, M simply connected. Let $\partial^\nabla: \Omega^1(M, E) \rightarrow \Omega^2(M, E)$ be the antisymmetrisation of $\nabla: \Omega^1(M, E) \rightarrow \Omega^1(M, E \otimes T^*M)$. Let $g \in \Omega^1(M, E)$.

Then $\partial^\nabla g = 0 \iff g = \partial^\nabla \alpha$

for some $\alpha \in \Omega^0(M, E)$.

Proposition (Dual flatness): (M, g, ∇, ∇^*) is dually flat if and only if ∇ is flat and M is **Hessian** w.r.t. ∇ , i.e. there is $\Psi \in C^\infty(M)$ such that

$$g(X, Y) = (\nabla_X \partial \Psi)(Y) =: \text{Hess}^\nabla(\Psi)(X, Y).$$

I $\Rightarrow$ II First, ∇ is torsion-free, since $0 = \partial^\nabla \Psi = g(X, Y) - g(Y, X) + \partial \Psi(T^2(X, Y))$. since g is non-degenerate

torsion-freeness and flatness of ∇^* follow from duality.

II $\Rightarrow$ I We have $g \in \Omega^1(M, T^*M)$ and $\partial^\nabla g(X, Y) = \nabla_X(g(Y)) - \nabla_Y(g(X)) - g([X, Y], \cdot)$. $\nabla_X Y - \nabla_Y X = [X, Y]$

$$= (\nabla_X g)(Y) - (\nabla_Y g)(X) = 0$$

since ∇g is Codazzi. Hence, by Poincaré's lemma, $g = \partial^\nabla \alpha$ for some $\alpha \in \Omega^0(M, T^*M) = \Omega^1(M)$, i.e. $g = \nabla \alpha$. Now

$$\partial^\nabla \alpha(X, Y) = (\nabla_X \alpha)(Y) - (\nabla_Y \alpha)(X) + \alpha(T^2(X, Y)) = 0,$$

hence $\alpha = \partial \Psi$ for some $\Psi \in \Omega^0(M) = C^\infty(M)$. □

Note: In other words, dually flat manifolds are manifolds with local coordinates θ with vanishing Christoffel symbols and such that

$$g_{ij} = \frac{\partial^2 \Psi}{\partial \theta_i \partial \theta_j}$$

for some $\Psi \in C^\infty(M)$.

Proposition (Pythagorean): Let M be dually flat. Let $P, Q, R \in M$ be connected by the ∇ -geodesic γ_{PQ} and the ∇^* -geodesic σ_{QR} , respectively. If γ_{PQ} and σ_{QR} intersect orthogonally at Q then

$$D_\Psi[P: R] = D_\Psi[P: Q] + D_\Psi[Q: R].$$

Using $D_\Psi[P: Q] = \Psi(\theta_P) + \Psi^*(\theta_Q^*) - \theta_P \cdot \theta_Q^*$, we have

$$D_\Psi[P: Q] + D_\Psi[Q: R] - D_\Psi[P: R] = (\theta_P - \theta_Q) \cdot (\theta_Q^* - \theta_R^*).$$

In affine coordinates, γ_{PQ} is proportional to $\theta_P - \theta_Q$ and σ_{QR} to $\theta_Q^* - \theta_R^*$, which are orthogonal. □

Note: We only consider manifolds covered by one (affine) coordinate chart. In particular, any two points are connected by a unique ∇ -geodesic.

Proposition (Projection): Let M be dually flat, $P \in M$, $S \subset M$ a submanifold. Then $R \in S$ is a critical point for $S \rightarrow \mathbb{R}, R \mapsto D_\Psi[P: R]$ if and only if the ∇ -geodesic from P to R is a projection, i.e. $T_R S$ is orthogonal to it.

If R is the projection then any ∇^* -geodesic at R in S gives a Pythagorean triangle, so R is a critical point by the Pythagorean theorem.

Conversely, we have

$$D(P: R) = \Psi(\theta_P) + \Psi^*(\theta_R^*) - \theta_P \cdot \theta_R^*,$$

so differentiating, we get

$$\frac{\partial}{\partial \theta_R^i} D(P: R) = \frac{\partial \Psi^*}{\partial \theta_R^i}(\theta_R^*) - \theta_{P,i} = \theta_{R,i} - \theta_{P,i}$$

Hence,

$$\frac{\partial}{\partial \theta_R} D(P: R(0)) = (\theta_P - \theta_R) \cdot \vec{R}(0).$$

This is zero for any $\vec{R}(0) \in T_R S$ if and only if $(\theta_P - \theta_R) \perp T_R S$. □

Corollary: If $S \subset M$ is flat then R minimises $D_\Psi[P: R]$ on S if and only if R is the ∇ -geodesic projection of P .

1) Exponential and Mixture families

An **exponential family of probability distributions** is a manifold of probability distributions parameterised by $\theta \in \mathbb{R}^n$ of the form

$$p(x, \theta) = \exp(\theta \cdot x + k(x) - \Psi(\theta)).$$

Recall that $\int p(x, \theta) dx = 1$ yields the convex function

$$\Psi(\theta) = \log \int \exp(\theta \cdot x) \exp(k(x)) dx,$$

giving M the structure of a dually flat manifold. Recall that

$$\eta := \theta^* = \nabla \Psi(\theta) = \mathbb{E}[x].$$

We denote $\varphi(\eta) = \Psi^*(\theta^*)$. Note that by linearity of expectations,

$$\mathbb{E}_\theta[\log p(x, \theta)] = \theta \cdot \eta - \Psi(\theta) + \mathbb{E}_\theta[k(x)].$$

We also have

$$\begin{aligned} g_{ij}(\theta) &= \partial_i \partial_j \Psi(\theta) = \mathbb{E}[(x_i - \eta_i)(x_j - \eta_j)] \\ &= \mathbb{E}[\partial_i \log p(x, \theta) \partial_j \log p(x, \theta)], \end{aligned}$$

the **Fisher information matrix**.

Example (Gaussian): Let

$$\begin{aligned} p(x, \mu, \sigma) &= \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \\ &= \exp\left(-\frac{1}{2} \log(2\pi\sigma) + \frac{\mu x}{\sigma^2} - \frac{x^2}{2\sigma^2} - \frac{\mu^2}{2\sigma^2}\right) \\ &= \exp\left(\underbrace{\frac{\mu}{\sigma^2}}_{\theta^1} x - \underbrace{\frac{1}{2\sigma^2}}_{\theta^2} x^2 - \left(\underbrace{\frac{\mu^2}{2\sigma^2} + \frac{1}{2} \log(2\pi\sigma)}_{\Psi}\right)\right). \end{aligned}$$

$x = (x_1, x_2)$

Also,

$$\Psi(\theta) = -\frac{(\theta^1)^2}{4\theta^2} + \frac{1}{2} \log\left(-\frac{\pi}{\theta^2}\right), \quad k(x) = 0,$$

and

$$\begin{aligned} \eta_1 &= \frac{\partial \Psi}{\partial \theta^1} = -\frac{\theta^1}{2\theta^2} = \mu = \mathbb{E}[X] \\ \eta_2 &= \frac{\partial \Psi}{\partial \theta^2} = \frac{(\theta^1)^2}{4(\theta^2)^2} - \frac{1}{2\theta^2} = \mu^2 + \sigma^2 = \mathbb{E}[X^2]. \end{aligned}$$

Example (Discrete): The manifold of discrete distributions on $n+1$ variables

$$p(x) = \sum p_i \delta_i(x)$$

is the n -simplex S_n . We have

$$\log p(x) = \sum \log \frac{p_i}{p_0} \delta_i(x) + \log p_0,$$

and

$$\Psi(\theta) = -\log p_0 = \log(1 + \sum \exp \theta^i).$$

The dual affine coordinates are

$$\eta_i = \mathbb{E}[x_i] = p_i$$

and

$$\varphi(\eta) = \sum \eta_i \log \eta_i + (1 - \sum \eta_i) \log(1 - \sum \eta_i),$$

giving

$$\theta^i = \nabla_i \varphi(\eta) = \log \frac{\eta_i}{1 - \sum \eta_i}.$$

Given $n+1$ probability distributions $q_0, \dots, q_n$ on the same space, we can form a **mixture family** parameterised by $\eta \in S_n$.

$$p(x, \eta) = \sum \eta_i q_i(x)$$

Note that

$$\varphi(\eta) = \int p(x, \eta) \log p(x, \eta) dx$$

is convex. We regard this as a dual convex function, giving $M = \{p(x, \eta)\}$ a dual affine coordinate system. The primal coordinates are $\theta = \nabla \varphi(\eta)$. In the case of S_n , this is the natural parameter for the exponential family.

Application (Maximum entropy): Let $c_i: \{0, \dots, n\} \rightarrow \mathbb{R}$ be k random variables, $a \in \mathbb{R}^k$. Let

$$S_n \supset M_{n,k}(a) =: \{p \in S_n \mid \mathbb{E}_p[c_i] = a_i\}.$$

This is a flat submanifold. Let $P_0 = \frac{1}{n+1}(1, \dots, 1) \in S_n$ be the uniform distribution. Note that it maximises entropy. We have

$$D_\varphi[P_0: P] = \Psi(\theta_0) + \varphi(\eta) - \theta_0 \cdot \eta = D_{KL}[P: P_0].$$

Hence, maximising the entropy in $M_{n,k}$ is equivalent to minimising $D_{KL}[P: P_0]$. By the projection theorem, the maximiser is the ∇ -geodesic projection of P_0 .

Application (Mutual information): Consider the manifold M of distributions on two random variables x, y , and the submanifold $M_I \subset M$ of independent distributions $p(x, y) = p(x)p(y)$. M_I is flat since the geodesics between two independent distributions is independent.

Given any $p \in M$, let $\hat{p} \in M_I$ be its ∇^* -projection, i.e. $\hat{p}$ minimises $M_I \ni p_I \mapsto D_{KL}(p: p_I)$. We have

$$\begin{aligned} D_{KL}(p: p_I) &= \int p(x, y) \log \frac{p(x, y)}{p_I(x, y)} \\ &= \int p(x, y) \log p(x, y) - \int p(x, y) \log p_I(x, y) \\ &\stackrel{p_I \text{ factors}}{=} (\text{const. in } p_I) - \int p(x, y) \log p_I(x) - \int p(x, y) \log p_I(y) \\ &= (\text{const.}) - \underbrace{\int \int p(x, y) \log p_I(x)}_{\text{marginal distribution of } p} - \int \int p(x, y) \log p_I(y) \end{aligned}$$

Since cross-entropy is maximised when the distributions match, $p_I(x)$ and $p_I(y)$ are the marginal distributions of p . Hence, the distance to the ∇^* -projection of p is the **mutual information**.

Application (Maximum Likelihood Estimator): Let $x_1, \dots, x_N$ be independent samples of $p(x, \theta)$ in an exponential family, used to estimate θ . Then the joint distribution

$$p(x_1, \dots, x_N, \theta) = \prod p(x_i, \theta) = \exp(N\theta \cdot \bar{x} - N\Psi(\theta)) =: p_N(\bar{x}, \theta).$$

So let $M = S_n$, $S = \{p(x, u)\}$, where $u \in \mathbb{R}^k$, a submanifold defining the statistical model. The samples specify a point $\bar{\eta} = \bar{x} \in M$, with $\bar{\theta}$ as θ -coordinates. Then

$$\bar{p}(x) = \frac{1}{N} \sum \delta(x - x_i)$$

and minimising the divergence

$$u \mapsto D_{KL}(\bar{p}(x): p(x, u))$$

is equivalent to minimising the log-likelihood

$$u \mapsto \sum \log p(x_i, u).$$

Hence, the **maximum likelihood estimator** is the ∇^* -projection of $\bar{p}$ to S .

2) Invariant geometry

Let $M = \{p(x, \gamma)\}$. What might a natural divergence on M look like?

Let $k: X \rightarrow Y$ be some map. Then $k: M \rightarrow M' = \{\bar{p}(\gamma, \gamma)\}$ by

$$\bar{p}(\gamma, \gamma) = \int_{k(x)=\gamma} p(x, \gamma).$$

Generally, this transformation loses information. We postulate that

$$\bar{D}[\gamma: \gamma'] \leq D[\gamma: \gamma'] \quad (*)$$

We call a function $s: X \rightarrow Y$ a **sufficient statistic** for M if

$$p(x, \gamma) = \bar{p}(s, \gamma) r(x).$$

We will require equality in $(*)$ to hold if and only if k is a sufficient statistic.

Generally, we say that a geometric structure on M , like a divergence or a metric, satisfies the **invariance criterion** if it satisfies a monotonicity like $(*)$ with equality for exactly the sufficient statistics.

Example (Coarse graining): Let $X = \{1, \dots, n\}$ be partitioned into $X = \bigsqcup_{i=0}^m X_i$. Consider a family S_n of discrete probability distributions. Let

$$\gamma: X \rightarrow \{0, \dots, m\}, \quad i \mapsto \delta(i \in X_j)$$

be the **coarse graining** statistic. If $p = (p_0, \dots, p_m)$ then $\bar{p} = (\bar{p}_0, \dots, \bar{p}_m)$ where

$$\bar{p}_a = \sum_{i \in X_a} p_i.$$

Let $D[p:q]$ be decomposable as

$$D[p:q] = \sum \Delta(p_i, q_i).$$

Then

$$\bar{D}[\bar{p}:\bar{q}] = \sum \Delta(\bar{p}_i, \bar{q}_i).$$

We have $p(x, \gamma) = p(\gamma, \gamma) p(x|\gamma, \gamma)$.

Hence, γ is a sufficient statistic if $p(x|\gamma, \gamma) = p(x|\gamma)$ does not depend on γ .

We call divergences of the form

$$D_f[p:q] = \sum p_i f\left(\frac{q_i}{p_i}\right)$$

f-divergences if f is differentiable, convex, and $f(1) = 0$.

Note: f-divergences are invariant and decomposable.

Note: If $n > 1$, any invariant decomposable divergence is an f-divergence.

Note: By adding a linear function, we can assume that $f(1) = 0 = f'(1)$. Since $D_{cf} = cD_f$, we can scale f so that $f''(1) = 1$. We call such f **standard**. Note that $f^*(u) = uf(\frac{1}{u})$, so the dual of f is also standard.

Example (α -divergence): Let $\alpha \in \mathbb{R} \setminus \pm 1$

$$f_\alpha(u) = \frac{4}{1-\alpha^2} \left(1 - u^{\frac{1+\alpha}{2}}\right).$$

Note that $D_\alpha[p:q] = D_{-\alpha}[q:p]$.

Taking the limits $\alpha \rightarrow \pm 1$, we get

$$f_\alpha(u) = \begin{cases} u \log u & \alpha = 1 \\ -\log u & \alpha = -1 \end{cases}$$

giving the KL-divergence and its dual at $-1, 1$, respectively.

Proposition: The following properties hold in S_n :

- $D_f[p, q]$ is convex in p and q ;
- $0 \leq D_f[p, q] \leq \begin{cases} \lim_{u \rightarrow 0} f(u) + uf(\frac{1}{u}) \\ \sum (p_i - q_i) f'(p_i/q_i) \end{cases}$

- For $\alpha \geq 1$, $D_\alpha[p, q] = \infty$

if there is $x \in X$ s.t. $p(x) = 0 \neq q(x)$, and dually for $\alpha \leq -1$.

Corollary: Let $p \in S_n$, $S \subset S_n$, $S \ni \hat{p}_S = \arg\min_{q \in S} D_\alpha[p:q]$. Then

- For $\alpha \geq 1$, $\hat{p}_S$ is zero wherever p is;
- For $\alpha \leq -1$, $\hat{p}_S$ is non-zero wherever p is.

Note: The Riemannian metric associated with any standard f-divergence is given by the **Fisher information matrix**

$$g_{ij} = \mathbb{E}[\partial_i \log p(x, \gamma) \partial_j \log p(x, \gamma)].$$

Conversely, the Fisher information matrix is the unique invariant metric on S_n , up to scaling.

$$u(\gamma) = \mathbb{E}[\hat{\gamma}] - \gamma \rightarrow 0 \text{ as } N \rightarrow \infty$$

Theorem (Cramér-Rao): Let $M = \{p(x, \gamma)\}$, $x_1, \dots, x_N$ sampled from $p(x, \gamma)$, $\hat{\gamma} = f(x_1, \dots, x_N)$ an estimator. If $\hat{\gamma}$ is asymptotically unbiased then

$$\mathbb{E}[(\hat{\gamma}_i - \gamma_i)(\hat{\gamma}_j - \gamma_j)] \geq \frac{1}{N} G^{-1}$$

accuracy

Fisher information matrix